STEELWORK NOTES

GENERAL

1. FABRICATE AND ERECT STRUCTURAL STEELWORK IN ACCORDANCE WITH AS 4100 AND AS/NZS 5131.
2. THESE DRAWINGS HAVE BEEN PREPARED TO INDICATE THE STRUCTURAL INTENT. THE SHOP DETAILER IS TO USE THESE DRAWINGS AS A BASIS FOR DIMENSIONAL COORDINATION WITH OTHER CONSULTANT'S DRAWINGS AND IS TO PREPARE DETAILED SHOP DRAWINGS. WHERE NECESSARY, THE SHOP DETAILER IS TO MAKE ASSUMPTIONS AND SUBMIT TO NORTHROP CONSULTING ENGINEERS FOR RESOLUTION. SHOP DETAILER IS TO ALLOW TO REWORK SHOP DRAWINGS AS NECESSARY. FABRICATOR SHALL PREPARE SHOP DRAWINGS AND SUBMIT THEM TO THE BUILDER FOR THEIR APPROVAL. **ALLOW 5 WORKING DAYS FOR THE REVIEW OF SHOP DRAWINGS.** NO STEELWORK SHALL BE FABRICATED UNTIL THE FINAL APPROVAL OF THE SHOP DRAWINGS HAVE BEEN APPROVED AND ALL COMMENTS HAVE BEEN RESOLVED.
3. THE FABRICATION AND ERECTION OF THE STRUCTURAL STEELWORK SHALL BE SUPERVISED BY A QUALIFIED PERSON EXPERIENCED IN SUCH SUPERVISION, IN ORDER TO ENSURE THAT ALL REQUIREMENTS OF THE DESIGN ARE MET. ALL WORK SHALL BE UNDERTAKEN BY COMPETENT PERSONNEL. REQUIREMENT AND EXAMPLES OF QUALIFICATIONS FOR COMPETENT PERSONNEL ARE CONTAINED IN AS/NZS 5131.
4. FABRICATION TOLERANCES SHALL CONFORM TO THE REQUIREMENTS OF AS/NZS 5131. THE TOLERANCE CLASS FOR FUNCTIONAL TOLERANCES SHALL BE CLASS 1 UNO FOR CONSTRUCTION CATEGORY CC1 AND CC2, AND CLASS 2 FOR CONSTRUCTION CATEGORY CC3 AND CC4.
5. THESE DRAWINGS MAY NOT IDENTIFY ALL SECONDARY STEELWORK ELEMENTS THAT ARE REQUIRED FOR SUPPORT, FIXING AND FINISHING OF GLAZING, CLADDING AND LINING. THE TENDERER IS RESPONSIBLE FOR THE INCLUSION OF SUCH STEELWORK ELEMENTS TO THE EXTENT REQUIRED ON THE ARCHITECT'S DRAWINGS. INCLUDING BUT NOT LIMITED TO HEAD RESTRAINT FOR INTERNAL WALLS, TRACK LIGHTING BARS, CURTAIN RODS, SUPPORT FOR MECHANICAL EQUIPMENT, WINDOW HEADS/SILLS, SUPPORT FOR ROLLER DOORS/SHUTTERS, SUPPORT FOR PROPRIETARY AWNINGS, HOODS AND SUN SCREENS/SHADES, AND SUPPORT FOR JOINERY ITEMS.
6. FIRE PROTECTION OF ALL STRUCTURAL STEEL ELEMENTS IS THE RESPONSIBILITY OF THE CONTRACTOR. SUITABLE PROVISION SHALL BE MADE TO ENSURE THAT THE FIRE RATING LEVELS REQUIRED BY THE NATIONAL CONSTRUCTION CODE (NCC) AND THE FIRE ENGINEER ARE ACHIEVED, INCLUDING ALLOWANCE FOR SUITABLE PROTECTIVE COATINGS AND CLADDING AS REQUIRED.
7. PROVIDE HOLES, CLEATS AND FIXING FOR LIGHT STEEL/TIMBER FRAMING, FINISHES, ETC. SHOWN ON ARCHITECTURAL DRAWINGS.
8. ALL MEMBERS SHALL BE SUPPLIED IN A SINGLE LENGTHS. SPLICES SHALL ONLY BE PERMITTED IN LOCATIONS SHOWN ON THE STRUCTURAL DRAWINGS. SPLICES AT OTHER LOCATIONS SHALL BE APPROVED BY NORTHROP PRIOR TO FABRICATION COMMENCING.

MATERIALS

1. ALL STRUCTURAL STEEL MATERIAL SHALL CONFORM TO THE FOLLOWING STANDARDS AND MINIMUM STRENGTH GRADES:
   - HOT ROLLED SECTIONS = AS/NZS 3679.1, TS102 300MPa
   - PLATE = AS/NZS 3678, TS102 250MPa
   - FLATS = AS/NZS 1594, TS102 250MPa
   - SQUARE HOLLOW SECTIONS = AS/NZS 1163, TS102 350MPa
   - RECTANGULAR HOLLOW SECTIONS = AS/NZS 1163, TS102 350MPa
   - CIRCULAR HOLLOW SECTIONS = AS/NZS 1163, TS102 350MPa
   - WELDED BEAMS AND COLUMNS = AS/NZS 3679.2, TS102 300MPa
   - SHEAR STUDS = AS/NZS 1554.2 380MPa
   - PURLINS AND GIRTS = AS 1397 450MPa
2. MEMBER SIZES SHALL BE AS SHOWN ON THE STRUCTURAL DRAWINGS. NO SUBSTITUTION IS PERMITTED WITHOUT APPROVAL IN WRITING FROM THE ENGINEER.
3. DOCUMENTATION SUPPLIED WITH MATERIALS AND COMPONENTS SHALL CONFORM TO THE REQUIREMENTS OF AS/NZS 5131.
4. PARTIAL TRACEABILITY SHALL BE PROVIDED FOR CONSTRUCTION CATEGORY CC2 AND FULL TRACEABILITY FOR CC3 AND CC4 IN ACCORDANCE WITH AS/NZS 5131.
5. WHERE STEEL ELEMENTS SHOWN ON THE STRUCTURAL OR ARCHITECTURAL DRAWINGS ARE REQUIRED TO BE CURVED, BENT, ROLLED OR PRECAMBERED, THE FABRICATION CONTRACTOR SHALL BE RESPONSIBLE FOR THE METHODS TO ACHIEVE THE REQUIRED SHAPES WITHOUT LOCALISED DISTORTION OR COMPROMISING THE STRENGTH OF THE MEMBERS.

MATERIAL SOURCES (AUSTRALIAN SUPPLIER)

1. ALL STRUCTURAL STEEL SHALL BE SOURCED FROM MILLS WITH A RELEVANT JAS ANZ ACCREDITED THIRD-PARTY CERTIFICATION SUCH AS THE ACRS SCHEME. ALTERNATIVE SOURCING OF THIRD-PARTY CERTIFIED STRUCTURAL STEEL SHALL BE SUBMITTED FOR REVIEW AND MUST BE APPROVED PRIOR TO THE COMMENCEMENT OF PROCUREMENT.
2. THE FOLLOWING REPORTS FROM TESTING ORGANISATIONS ACCREDITED BY A TESTING AUTHORITY RECOGNISED IN AUSRALIA (E.G. NATA OR JAS-ANZ ACCREDITED) ARE TO BE SUBMITTED TO NORTHROP CONSULTING ENGINEERS FOR APPROVAL PRIOR TO FABRICATION:
   - CERTIFIED MILL TEST REPORTS OR TEST CERTIFICATES SHALL BE PROVIDED AS EVIDENCE OF
   - COMPLIANCE WITH THE STANDARDS REFERRED TO IN AS4100.
   - TEST CERTIFICATES OF COMPLIANCE FOR ALL FASTENERS.
   - FOR COLD FORMED SECTIONS A "CERTIFICATE OF CONFORMITY TO AS1397".
3. UNIDENTIFIED STEEL (I.E. ANY STEEL THAT IS NOT ACCOMPANIED WITH EVIDENCE STATING COMPLIANCE WITH THE REQUIREMENT OF AS4100) SHALL ONLY BE USED STRICTLY IN ACCORDANCE WITH CLAUSE 2.2.3 OF AS4100.
4. IF MATERIALS SUPPLIED AND INSTALLED ARE SUBSEQUENTLY PROVEN TO BE NON COMPLIANT WITH THE SPECIFIED AUSTRALIAN STANDARDS IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY AND COST TO UNDERTAKE NATA OR EQUIVALENT CERTIFIED TESTING TO PROVE CONFORMANCE TO THE AUSTRALIAN STANDARDS AND DESIGN SPECIFICATIONS. SIMILARLY ANY RECTIFICATION WORKS THAT MAY SUBSEQUENTLY BE REQUIRED TO SATISFY AUSTRALIAN CODE REQUIREMENT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR.

MATERIAL SOURCES (IMPORTED SUPPLIER)

1. ALL IMPORTED STRUCTURAL STEELWORK USED ON THIS PROJECT SHALL BE COMPLIANT WITH AS4100, AND IN PARTICULAR:
   A. CERTIFIED MILL TEST REPORTS, OR TEST CERTIFICATES SHALL BE PROVIDED AS EVIDENCE OF COMPLIANCE WITH THE STANDARDS REFERRED TO IN AS4100. THESE CERTIFICATES SHALL BE SUBMITTED TO NORTHROP CONSULTING ENGINEERS FOR APPROVAL PRIOR TO COMMENCEMENT OF FABRICATION.
   B. PROVIDE TEST CERTIFICATED FOR COMPLIANCE FOR ALL FASTENERS. THESE CERTIFICATES SHALL BE SUBMITTED TO NORTHROP CONSULTING ENGINEERS FOR APPROVAL PRIOR TO FABRICATION.
   C. 'FOR COLD FORMED SECTIONS A "CERTIFICATE OF CONFORMITY TO AS1163-1991" SHALL BE SUBMITTED TO NORTHROP CONSULTING ENGINEERS FOR APPROVAL PRIOR TO FABRICATION.
   D. CERTIFICATES SHALL ONLY BE ACCEPTED FROM TESTING COMPANIES ACCREDITED BY A TESTING AUTHORITY RECOGNIZED IN AUSTRALIA, EG NATA OR JAS-ANZ CERTIFIED.
   E. UNIDENTIFIED STEEL EG. ANY STEEL THAT IS NOT ACCOMPANIED WITH EVIDENCE STATING COMPLIANCE WITH THE REQUIREMENT OF AS4100 SHALL ONLY BE USED STRICTLY IN ACCORDANCE WITH CLAUSE 2.2.3 OF AS4100.
   F. ALL BOLTS AND FITMENTS TO STEEL ARE TO BE LOCALLY MANUFACTURED AND CERTIFIED TO RELEVANT AUSTRALIAN STANDARDS.
   G. BUTT WELDS IN FLANGE OR WEB COMPONENTS ARE PERMITTED IF THE REQUIRED LENGTH OF THE MEMBER IS GREATER THAN THE AVAILABLE ROLLED LENGTH OF THE COMPONENT. THE JOINTS SHALL BE COMPLETE PENETRATION BUTT WELDS, GROUND FLUSH IN THE LONGITUDINAL DIRECTION OF THE SECTION AND SUBJECT TO 100% ULTRASONIC TESTING. BUTT JOINTS IN ANY COMPONENT SHALL BE COMPLETED BEFORE THE COMPONENT IS WELDED TO ANOTHER COMPONENT OF THE MEMBER IN ACCORDANCE WITH AS/NZ3679.2 CLAUSE 5.3.4
   H. WEB TO FLANGE WELD REQUIREMENTS TO BE IN ACCORDANCE WITH AS/NZ3679.2 CLAUSE 5.3.3 TABLE 1 JOINS TO BE SUB-ARC MACHINE WELDED. WELD CAPACITY TO EXCEED WEB CAPACITY IN TENSILE TEST. MINIMUM TEST FAILURE LOAD TO BE IN ACCORDANCE WITH AS3679.2 TABLE 1
   I. AUSTRALIAN STEEL INSTITUTE TN015 TESTING REQUIREMENTS TO BE USED FOR CONFORMITY ASSESSMENT.

STEELWORK NOTES CONTINUED

MATERIAL SOURCES (IMPORTED SUPPLIER) CONTINUED

2. IF MATERIALS SUPPLIED AND INSTALLED ARE SUBSEQUENTLY PROVEN TO BE NON COMPLIANT WITH THE SPECIFIED AUSTRALIAN STANDARDS IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY AND COST TO UNDERTAKE NATA OR EQUIVALENT CERTIFIED TESTING TO PROVE CONFORMANCE TO THE AUSTRALIAN STANDARDS AND DESIGN SPECIFICATIONS. SIMILARLY ANY RECTIFICATION WORKS THAT MAY SUBSEQUENTLY BE REQUIRED TO SATISFY AUSTRALIAN CODE REQUIREMENT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR.

CONNECTIONS

1. ALL CUTTING, HOLING AND SHAPING OF STRUCTURAL STEEL SHALL CONFORM TO THE REQUIREMENTS OF AS/NZS 5131. PENETRATIONS OR CUT-OUTS OTHER THAN THOSE SHOWN ON THE DRAWINGS SHALL NOT BE MADE WITHOUT PRIOR APPROVAL.
2. ALL DETAILING NOT SPECIFICALLY SHOWN SHALL BE IN ACCORDANCE WITH THE AUSTRALIAN STEEL INSTITUTE (ASI) CURRENT EDITIONS OF THE STANDARDISED STRUCTURAL CONNECTIONS.
3. UNLESS NOTED OTHERWISE:
   A. ALL WELDS SHALL BE 6mm CONTINUOUS FILLET WELD (CFW) ALL ROUND
   B. ALL STEEL TO STEEL BOLTED CONNECTIONS SHALL BE MINIMUM TWO M20 GRADE 8.8/S
   C. ALL BASE CONNECTIONS SHALL HAVE MINIMUM FOUR M20 GRADE 4.6/S HOLD DOWN BOLTS
   D. A MINIMUM OF TWO THREADS SHALL EXTEND PAST THE NUT
   E. ALL PLATES SHALL BE 10mm THICK
   F. ALL PURLIN CLEATS SHALL BE 8mm THICK
4. THE ENDS OF HOLLOW SECTION MEMBERS SHALL BE SEALED WITH NOMINAL THICKNESS PLATES AND CONTINUOUS SEAL WELDED UNLESS NOTED OTHERWISE. IF HOLLOW SECTIONS ARE TO BE HOT-DIP GALVANIZED, VENT AND DRAINAGE HOLES SHALL BE PROVIDED CONFORMING TO THE REQUIREMENTS OF AS/NZS 5131 IN NON-VIEWABLE LOCATIONS.

BOLTING

1. ALL BOLTS ARE TO BE INSTALLED WITH ONE HARDENED WASHER UNDER THE TURNED PART AND WITH A MINIMUM OF TWO THREADS EXTENDING PAST THE NUT.
2. ALL BOLTS ASSEMBLIES, HOLD DOWN BOLTS AND MASONRY ANCHORS SHALL BE HOT DIP GALVANISED TO AS1214. ALL GALVANISED COMPONENTS TO BE CAST INTO CONCRETE MUST BE PASSIVATED.
3. NO CONNECTION SHALL HAVE LESS THAN 2 BOLTS UNLESS EXPLICITLY NOTED OTHERWISE.
4. BOLT SYMBOLS:
   - 4.6/S = GRADE 4.6 BOLT TO AS 1111 / SNUG TIGHTENED.
   - 8.8/S = GRADE 8.8 BOLT TO AS/NZS 1252.1 / SNUG TIGHTENED.
   - 8.8/TB = GRADE 8.8 BOLT TO AS/NZS 1252.1 / FULLY TENSIONED FOR BEARING JOINTS.
   - 8.8/TF = GRADE 8.8 BOLT TO AS/NZS 1252.1 / FULLY TENSIONED FOR FRICTION JOINTS.
   - 10.9/TB = GRADE 10.9 BOLT TO AS/NZS 1252.1 / FULLY TENSIONED FOR BEARING JOINTS.
   - 10.9/TF = GRADE 10.9 BOLT TO AS/NZS 1252.1 / FULLY TENSIONED FOR FRICTION JOINTS.
5. HIGH STRENGTH STRUCTURAL BOLTS SHALL BE VERIFIED TO AS/NZS 1252.2. THE DOCUMENTATION REQUIRED BY THE STANDARD, INCLUDING THE 'SUPPLIER DECLARATION OF CONFORMITY' (SDoC) SHALL BE PROVIDED.
6. HIGH STRENGTH BOLTS (I.E. GRADE 8.8 AND ABOVE) ARE NOT TO BE WELDED.
7. ALL SNUG TIGHTENING AND TENSIONING OF BOLTS ARE TO BE CARRIED OUT IN ACCORDANCE WITH AS/NZS 5131.
8. LOAD INDICATING WASHERS SHALL BE USED FOR ALL FULLY TENSIONED BOLTS (8.8/TB, 8.8/TF, 10.9/TB AND 10.9/TF). THE BOLTS SHALL BE INSTALLED USING THE DIRECT TENSION INDICATOR METHOD TO AS/NZS 5131. WHERE THE PART TURN METHOD IS TO BE USED, THE CALIBRATION METHODOLOGY IS TO BE SUBMITTED TO NORTHROP CONSULTING ENGINEERS FOR APPROVAL.
9. FOR CONNECTIONS WHERE 8.8/TF OR 10.9/TF BOLTS ARE SPECIFIED, A FRICTION COEFFICIENT OF 0.35 HAS BEEN ASSUMED FOR THE DESIGN. THE CONTACT SURFACES SHALL BE PREPARED ACCORDING TO AS/NZS 5131 AND BE FREE FROM PAINT, LACQUER OR OTHER APPLIED FINISHES.

WELDING

1. WELDING CONSUMABLES SHALL CONFORM TO THE REQUIREMENTS OF AS/NZS1554. WHERE THE YIELD STRENGTH OF STEEL TO BE WELDED IS LESS THAN OR EQUAL TO 500MPa, WELDING CONSUMABLES SHALL COMPLY WITH AS/NZS 1554.1, AND WHERE THE YIELD STRENGTH OF STEEL TO BE WELDED IS GREATER THAN 500MPa AND LESS THAN 690MPa WELDING CONSUMABLES SHALL COMPLY WITH AS/NZS 1554.4.
2. THE NOMINAL WELD CONSUMABLE TENSILE STRENGTH SHALL BE 490MPa UNO.
3. ALL SHOP WELDS SHALL BE SP CATEGORY AND ALL SITE WELDS SHALL BE GP CATEGORY IN ACCORDANCE WITH AS/NZS 1554.1 AND AS/NZS 1554.4 UNO.
4. OTHER THAN THE SITE WELDS (IF ANY) SHOWN ON THE SHOP DRAWINGS, SITE WELDING IS NOT PERMITTED WITHOUT PRIOR APPROVAL.
5. ALL BUTT WELDS ARE TO BE COMPLETE PENETRATION BUTT WELDS UNO.
6. ALL SITE BUTT WELDS ARE TO BE SP CATEGORY AND SUBJECT TO VISUAL EXAMINATION, ULTRASONIC EXAMINATION AND MAGNETIC PARTICLE EXAMINATION.
7. THE EXTENT ON NON-DESTRUCTIVE WELD EXAMINATION IN ACCORDANCE WITH AS/NZS 5131 SHALL BE AS NOTED BELOW. RADIOGRAPHIC OR ULTRASONIC EXAMINATION SHALL BE TO AS/NZS 1554.1, AS 2177 AND AS2207 AS APPROPRIATE.

WELD CATEGORY	VISUAL SCANNING	VISUAL EXAMINATION	RADIOGRAPHY OR ULTRASONICS
GP	100	25	NIL
SP	100	50	10

LAMELLAR TEARING

1. JOINT DETAILS WHICH ARE SUSCEPTIBLE TO LAMELLAR TEARING (LT) ARE INDICATED ON THE PROJECT DRAWINGS AS 'LT SUSCEPTIBLE' AND INCLUDE ANY END PLATE WITH THICKNESS GREATER THAN 16mm. THE SPECIFIC PLATE IN JOINTS THAT ARE INDICATED AS 'LT SUSCEPTIBLE' SHALL BE SUPPLIED ULTRASONICALLY TESTED TO AS 1710 CLASS 1. JOINTS THAT ARE DESIGNATED 'LT SUSCEPTIBLE' AND FURTHER REQUIRE PLATE TO A NOMINATED Z-VALUE ARE INDICATED ON THE PROJECT DRAWINGS WITH A DESIGNATED Z-VALUE. THE PLATE USED FOR THESE JOINTS SHALL BE ORDERED TO THE DESIGNATED Z-VALUE.
2. SUPPLEMENTARY ULTRASONIC TESTING TO AS 2207 AND AS/NZS 1554.1 IS REQUIRED FOR ALL PLATES 40mm THICKNESS AND OVER.

PURLINS AND GIRTS

1. PURLINS AND GIRTS SHALL CONFORM TO THE REQUIREMENTS OF AS/NZS 5131.
2. THE SUSPENSION OF CEILINGS, SERVICES ETC FROM PURLINS AND GIRTS SHALL BE IN ACCORDANCE WITH THE APPROVED METHODS DETAILED BELOW;

3. UNLESS OTHERWISE DETAILED, FABRICATOR SHALL ALLOW TO PROVIDE ONE ROW OF BRIDGING FOR PURLINS SPANNING UP TO 4.0m, TWO ROWS FOR PURLINS SPANNING BETWEEN 4.0m AND 8.0m AND THREE ROWS FOR SPANS GREATER THAN 8.0m INSTALLED IN ACCORDANCE WITH MANUFACTURER'S WRITTEN INSTRUCTIONS.
4. PURLIN CLEATS SHALL GENERALLY BE MINIMUM 75 WIDE x 8 PLATE U.N.O. WHERE THE DISTANCE BETWEEN THE UNDERSIDE OF PURLIN AND THE TOP OF THE BEAM EXCEEDS 150mm THE CLEATS SHALL BE MINIMUM 75 x 6 EA. UP TO A MAXIMUM HEIGHT OF 800mm.
5. PURLINS SHALL BE LAPPED IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS UNLESS SHOWN ON THE DRAWINGS.

STEELWORK NOTES CONTINUED

PURLINS AND GIRTS CONTINUED

6. UNLESS OTHERWISE DETAILED, FABRICATOR SHALL PROVIDE C10015 TRIMMER PURLINS ALONG WITH GENERAL PURPOSE BRACKETS TO SUPPORT EDGE OF ROOF SHEETING AT ALL HIPS, VALLEYS AND PENETRATIONS.
7. FIX CROSS BRACING TO PURLINS AT 3000 MAX CENTRES WITH M10 BOLTS OR M6 HOOKS. BOLTS ARE TO BE MECHANICALLY FIXED TO PURLINS, WITH HOOKS TO BE LOOPED THROUGH THE WEB OF THE PURLINS AND SECURED USING A NUT TO ENSURE THEY WILL NOT BECOME LOOSE DURING MOVEMENT OF THE ROOF. ALTERNATIVELY, GALVANISED STRAP MAY BE USED TO SECURE THE BRACING. THE STRAP IS TO BE INSTALLED TO LIFT THE BRACE SUCH THAT IT IS PLUMB, WITH THE BRACE SCREW FIXED TO THE WEB OF THE PURLIN AT BOTH ENDS USING 2/12-14g TEKS SCREWS.

SURFACE TREATMENTS

1. UNLESS NOTED OTHERWISE IN THE CONTRACTUAL DOCUMENTATION, THE MINIMUM SURFACE TREATMENT OF BOTH INTERNAL AND EXTERNAL STEELWORK SHALL CONFORM TO THE REQUIREMENTS OF AS/NZS 5131.
2. STRUCTURAL STEELWORK TO BE GALVANISED SHALL CONFORM TO THE REQUIREMENTS OF AS/NZS 5131 AND AS/NZS 4680. WHERE INTENDED TO BE GALVANISED, ENSURE STRUCTURAL STEELWORK IS DETAILED IN ACCORDANCE WITH AVAILABLE GALVANISING BATH RESTRICTIONS.
3. ALL PAINTED SURFACES SHALL ACHIEVE TREATMENT GRADE P2 AND ALL SURFACES DESIGNED AS ARCHITECTURAL EXPOSED STRUCTURAL STEELWORK SHALL BE PREPARED TO ACHIEVE TREATMENT GRADE P3 IN ACCORDANCE WITH AS/NZS 5131.
4. ALL PAINTED INDOOR STEELWORK PROTECTED FROM WEATHER IS TO ACHIEVE COATING QUALITY LEVEL PC1 AND ALL OTHER PAINTED STEELWORK IS TO ACHIEVE COATING QUALITY LEVEL PC2 IN ACCORDANCE WITH AS/NZS 5131.
5. ALL BURIED STEELWORK IS TO BE PAINTED WITH 2 COATS OF BITUMINOUS PAINT IN ADDITION TO THE 'EXPOSED TO WEATHER' COATING NOMINATED BELOW OR ENCASED IN CONCRETE WITH MINIMUM 75mm COVER TO CONCRETE.
6. ALL CONCRETE ENCASED STEELWORK SHALL BE UNPAINTED, FREE OF LOOSE RUST, LOOSE MILL SCALE, DIRT, OIL, GREASE, ETC. STEEL IS TO ACHIEVE 75mm MINIMUM COVER TO CONCRETE AND CONCRETE ENCASEMENT IS TO BE REINFORCED WITH SL41 MESH UNO.
7. MEMBERS TO BE FIRE SPRAYED MUST NOT BE PAINTED AND MUST BE FREE OF SCALE. MEMBERS TO BE PAINTED WITH INTUMESCENT PAINT ARE TO BE PREPARED IN ACCORDANCE WITH THE PAINT MANUFACTURER'S SPECIFICATIONS. FOR MEMBERS TO BE PROTECTED WITH FIRE RATED PLASTERBOARD, INSTALLATION AND JOINTING OF THE PLASTERBOARD TO ARE TO BE AS PER THE ARCHITECT'S DETAILS AND MANUFACTURER'S SPECIFICATIONS.
8. ARCHITECTURALLY EXPOSED STRUCTURAL STEEL (AESS) SHALL CONFORM TO THE REQUIREMENTS OF AS/NZS 5131. AREAS TO BE TREATED AS AESS AND THE AESS CATEGORY (1, 2, 3, 4 OR C) ARE NOTED ON THE DRAWINGS AS AESS-1, ETC. IF NO CATEGORY IS PROVIDED, AESS COMPONENTS SHALL BE AESS 2. THE FABRICATOR SHALL REVIEW THE ARCHITECTURAL DRAWINGS TO DETERMINE IF ANY STEELWORK NOT NOMINATED ON THE DRAWINGS AS AESS REQUIRE AESS TREATMENT GIVEN THEIR VISIBILITY.
9. A CORROSIVITY CATEGORY OF **C3** HAS BEEN DETERMINED FOR THIS SITE. THE FOLLOWING SURFACE TREATMENTS HAVE BEEN SELECTED FROM AS/NZS 2312.1 AND AS/NZS 2312.2 AS APPROPRIATE TO ACHIEVE A MINIMUM 25 YEARS TO FIRST MAINTENANCE. ALL PAINT SYSTEM DESIGNATIONS ARE AS PER AS/NZS 2312.1 AND 'HDG' REFERS TO HOT DIP GALVANISING IN ACCORDANCE WITH AS/NZS 4680 AND AS 1397 AS APPROPRIATE. WHERE EXTERNAL COATINGS ARE NOTED AS NOT PERMITTED AS A METHOD OF CORROSION PROTECTION GRADE 316 STAINLESS STEEL IS TO BE USED IN LIEU OF STRUCTURAL STEELWORK.

CORROSIVITY CATEGORY **C3** : 1-50km FROM COAST, NO SALT SPRAY:

- PROTECTED FROM WEATHER	IZS2
- EXPOSED TO WEATHER	HDG-600, PUR5 OR IZS3
- BUILT INTO THE INTERNAL SKIN OF EXTERNAL WALLS	HDG-600, PUR5 OR IZS3
- PURLINS, GIRTS, AND ACCESSORIES	Z350

10. REFER TO ARCHITECTURAL DRAWINGS AND SPECIFICATION FOR COLOUR AND FINISH REQUIREMENTS FOR SURFACE TREATMENTS. WHERE ALTERNATIVES TO THE ABOVE SPECIFICATIONS ARE REQUIRED THESE ALTERNATIVES ARE ALL TO ACHIEVE A MINIMUM 25 YEARS TO FIRST MAINTENANCE AND ARE TO BE SUBMITTED TO NORTHROP CONSULTING ENGINEERS FOR APPROVAL.

ERECTION

1. STRUCTURAL STEELWORK ERECTION SHALL CONFORM TO THE REQUIREMENTS OF AS/NZS 5131.
2. THE STRUCTURAL STEELWORK ERECTOR SHALL BE RESPONSIBLE FOR TEMPORARY STABILITY DURING ERECTION AND SHALL PROVIDE AND LEAVE IN PLACE TEMPORARY BRACING AS IS NECESSARY TO SECURELY STABILISE THE STRUCTURE DURING ERECTION UNTIL NORTHROP CONSULTING ENGINEERS APPROVES THE REMOVAL OF THIS TEMPORARY BRACING.
3. PROVIDE AN ERECTION SEQUENCE METHODOLOGY FOR APPROVAL BY NORTHROP CONSULTING ENGINEERS PRIOR TO COMMENCING WITH ERECTION.
4. ALL MEMBERS HAVING A NATURAL CAMBER WITHIN THE STRAIGHTNESS TOLERANCE SHALL BE ERECTED WITH THE NATURAL CAMBER UP.
5. ERECTION TOLERANCE SHALL CONFORM TO THE REQUIREMENTS OF AS/NZS 5131. THE TOLERANCE CLASS FOR FUNCTIONAL TOLERANCES SHALL BE CLASS 1 FOR CONSTRUCTION CATEGORY CC2 AND CLASS 2 FOR CC3 AND CC4.
6. POST-INSTALLED MECHANICAL AND CHEMICAL ANCHORS AND THEIR INSTALLATION SHALL CONFORM TO THE REQUIREMENTS OF AS/NZS 5131 AND AS 5216. SITE TESTING SHALL BE PERFORMED ON MECHANICAL AND CHEMICAL ANCHORS TO VALIDATE CORRECT INSTALLATION, WITH AT LEAST 1 BOLT IN EACH CONNECTION TESTED OR AS OTHERWISE APPROVED BY NORTHROP CONSULTING ENGINEERS. SITE TESTING SHALL BE UNDERTAKEN ACCORDING TO THE REQUIREMENTS OF THE AEFAC TECHNICAL
   NOTE – SITE TESTING GUIDELINES VOLUMES 1 TO 4. PROVIDE ALL TEST RESULTS TO NORTHROP AND NOTIFY NORTHROP IMMEDIATELY IF A SINGLE FAILURE IS RECORDED. DO NOT PROCEED WITH INSTALLATION IF A FAILURE IS RECORDED.
7. ALL POST-INSTALLED MECHNICAL AND CHEMICAL ANCHORS ARE TO ACHIEVE A MINIMUM 100mm EDGE DISTANCE TO ANY CONCRETE OR MASONRY EDGE UNO.
8. GROUT ALL STEEL BASE PLATES THAT BEAR ON CONCRETE OR MASONRY IN ACCORDANCE WITH AS/NZS 5131 WITH NON-SHRINK GROUT THAT HAS A MINIMUM COMPRESSIVE STRENGTH OF 40MPa AT 7 DAYS.
9. PROPRIETARY ITEMS (E.G. PURLINS, ROOF/WALL SHEETING, FERRULES) SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S SPECIFICATIONS.
10. INSPECTION, TESTING, AND CORRECTION SHALL CONFORM TO THE REQUIREMENTS OF AS/NZS 5131.

STEELWORK NOTES CONTINUED

CHEMICAL ANCHORS

1. ALL BOLTED CONNECTIONS TO CONCRETE OR MASONRY ARE TO BE BY CHEMICAL INJECTION TYPE ANCHORS U.N.O.
2. UNLESS NOTED OTHERWISE ALL CHEMICAL ANCHORS SHALL BE HILTI HIT-HY 200 + HIT-V CHEMICAL INJECTION ANCHOR OR RAMSET CHEMSET REO 502 CHEMICAL INJECTION ANCHOR WITH GALVANISED GRADE 5.8 ANCHOR STUDS.
3. ALL ANCHORS TO BE INSTALLED BY SUITABLE QUALIFIED PERSON IN STRICT CONFORMANCE WITH MANUFACTURERS SPECIFICATIONS AND PROCEDURES
4. ALTERNATIVE CHEMICAL ANCHORS SYSTEMS ARE TO BE SUBMITTED TO NORTHROP ENGINEERS FOR REVIEW AND APPROVAL PRIOR TO USE.
5. SHOP DETAILER IS TO CONFIRM MINIMUM EDGE DISTANCE AND SPACING ARE ACHIEVED AS SET OUT IN THESE DOCUMENTS. IF MINIMUM DISTANCES ARE NOT ACHIEVABLE NOTIFY NORTHROP CONSULTING.
6. CLEAN AND DEGREASE BOLTS/ BARS PRIOR TO INSTALLATION
7. DIAMETER OF HOLES TO MANUFACTURERS SPECIFICATION FOR NOMINATED BOLT/ BAR DIAMETER.
8. DRILL HOLES USING ROTARY PERCUSSION DRILL BIT. DO NOT CORE HOLE FOR CHEMICAL ANCHORS.
9. ENSURE CHEMICAL IS ALLOWED TO FULLY CURE IN ACCORDANCE WITH MANUFACTURERS DETAILS PRIOR TO LOADING BOLTS/ BARS
10. CHEMICAL ANCHORS ARE TO BE TESTED BY A NATA REGISTERED LABORATORY TO 100% OF THE DESIGN LOAD NOTED BELOW IN ACCORDANCE WITH THE FOLLOWING REGIME:
    A. A MINIMUM OF 10% OF THE TOTAL NUMBER OR 5 (WHICH EVER IS GREATER) OF ANCHORS INSTALLED
    B. SHOULD 1 ANCHOR FAIL THEN AN ADDITIONAL 10% ARE TO BE TESTED.
    C. SHOULD A FURTHER ANCHOR FAIL THEN ALL ANCHORS ARE TO BE TESTED.
    D. ALL FAILED ANCHORS TO BE REMOVED AND REINSTALLED IN ACCORDANCE WITH THE MANUFACTURERS SPECIFICATIONS. ALL REINSTATED ANCHORS TO BE TESTED.
11. INSTALLATION SPECIFICATIONS TO MANUFACTURERS REQUIREMENTS. TYPICAL DESIGN INFORMATION:

ANCHOR SIZE	DRILLED HOLE	EMBEDMENT	TEST LOAD
M12	14	110	30kN
M16	18	125	50kN
M20	24	150	65kN

STRUCTURAL MODEL THEORETICAL EQUIVALENT EMBODIED CO^2 CONTENT

GENERAL

1. THE THEORETICAL EMBODIED CO^2 CONTENT HAS BEEN CALCULATED FROM VOLUMES DERIVED FROM THE STRUCTURAL MODEL AND GWP-GHG (A1-A3) SUMMARIES SOURCED FROM LOCAL SUPPLIERS (REFER TABLE 1).
2. VOLUMES ARE LIMITED TO PRIMARY AND SECONDARY STRUCTURAL ELEMENTS AS MODELLED IN THE STRUCTURAL REVIT MODEL. TERTIARY ELEMENTS SUCH AS TRIMMERS AND FLY BRACES ARE NOT INCLUDED.
3. REINFORCEMENT AND POST TENSIONING HAS NOT BEEN MODELLED BUT HAS BEEN INCLUDED IN THE EMBODIED CO^2 BASED ON RATES AS SPECIFIED IN TABLE 2 BELOW.
4. THE RATES PROVIDED IN TABLE 2 ARE FOR THE PURPOSES OF CALCULATING EMBODIED CO^2 AND SHOULD NOT BE USED FOR BUILDING ESTIMATION WITHOUT WRITTEN PERMISSION FROM NORTHROP CONSULTING.
5. TRANSPORTATION HAS NOT BEEN INCLUDED IN THE EMBODIED CO^2 CALCULATIONS.
6. THE SUMMARY OF A1-A3 EMBODIED CO^2 IS A REPRESENTATION OF THE CURRENT PHASE OR STAGE OF THE DESIGN MODEL.

TABLE 1 – GWP (A1-A3) PRODUCTION			
CONCRETE	kg $CO^2 eq/m^3$	CONCRETE	kg $CO^2 eq/m^3$
10MPa	250	40MPa	357
20MPa	271	50MPa	454
25MPa	288	60MPa	414
32MPa	314	80MPa	447
PRECAST CONCRETE		209	
TIMBER		kg $CO^2 eq/m^3$	
SOFT WOOD KD		-718	
HARD WOOD KD		-731	
GLULAM SOFT WOOD		-612	
GLULAM HARD WOOD		-408	
CLT PANEL		-492	
METALS		kg $CO^2 eq/t$	
REINFORCEMENT MESH		2060	
REINFORCEMENT BAR		1670	
REINFORCEMENT FIBRE		942	
REINFORCEMENT PRECAST		1840	
POST TENSIONING		3200	
HOT ROLLED STEEL		3320	
COLD FORMED STEEL		1900	
SHEETING		kg $CO^2 eq/m^2$	
PLY WOOD		-2.3	
METAL DECKING (BONDEK)		23.7	

TABLE 2 – CO^2 REINFORCEMENT RATES kg/m³			
FOOTINGS	150		
FOOTINGS FIBRE	25		
GROUND SLABS	115		
GROUND SLABS FIBRE	35		
STAIRS ON GROUND	80		
COLUMNS	300		
BLOCK WALLS	80		
PRECAST WALLS	110		
CONCRETE WALLS	150		
SUSPENDED SLAB REINF	130		
SUSPENDED BEAM REINF	230		
SUSPENDED PRECAST SLAB	125		
SUSPENDED SLAB PT	7	SLAB REINF	40
SUSPENDED BEAM PT	30	BEAM REINF	20
SUSPENDED COMPOSITE SLAB	120		
STAIRS	110		

SUMMARY OF A1-A3 EMBODIED CO^2 CONTENT FOR PRIMARY AND SECONDARY STRUCTURAL ELEMENTS			
CONCRETE COMBINED	kg CO^2 eq	HOT ROLLED STEEL	kg CO^2 eq
REINFORCEMENT	kg CO^2 eq	COLD FORMED STEEL	kg CO^2 eq
POST TENSIONING	kg CO^2 eq	TIMBER	kg CO^2 eq
COMPOSITE DECKING	kg CO^2 eq	STRUCTURAL PLYWOOD	kg CO^2 eq
GRAND TOTAL			kg CO^2 eq

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SKYGATE PLAY

STRUCTURAL SPECIFICATIONS SHEET 2

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